

第5章

5-3. $V = 1.0 \times 10^{-3} \text{ m}^3$, He. $m_1 = 4.0 \times 10^{-5} \text{ kg}$ H_2 . $m_2 = 4.0 \times 10^{-5} \text{ kg}$ $t = 30^\circ\text{C}$.

解: $p_{\Sigma} = p_1 + p_2$ ①

$p_1 V = \frac{m_1}{M_1} R T$ ②

$p_2 V = \frac{m_2}{M_2} R T$ ③

$T = 273 + t$ ④

$p_1 = 2.52 \times 10^4 \text{ Pa}$

$\Rightarrow p_2 = 5.04 \times 10^4 \text{ Pa}$

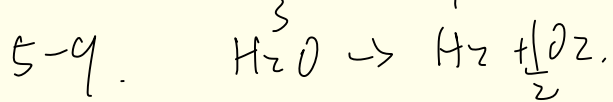
$p_{\Sigma} = 7.56 \times 10^4 \text{ Pa}$

$p_1 = \frac{4 \times 10^{-5} \times 8.31 \times 303}{1 \times 10^{-3}} = 20 \times 303 \times 8.31 \approx$

5-8.

解: $\frac{i}{2} \cdot \frac{m}{M} R \Delta T = 2.09 \times 10^2$ ① $\Rightarrow i = \frac{2Q}{R \Delta T} = 5$ $\Rightarrow$ 分子是双原子分子

$n = \frac{m}{M}$ ②



$$\frac{1}{2} \cdot \frac{M_3}{M_3} RT = Q_3 \quad (1)$$

$$\frac{1}{2} \cdot \frac{M_1}{M_1} RT = Q_1 \quad (2)$$

$$\frac{1}{2} \times \frac{1}{2} RT = Q_2 \quad (3)$$

$$\Delta Q = Q_1 + Q_2 - Q_3 \quad (4)$$

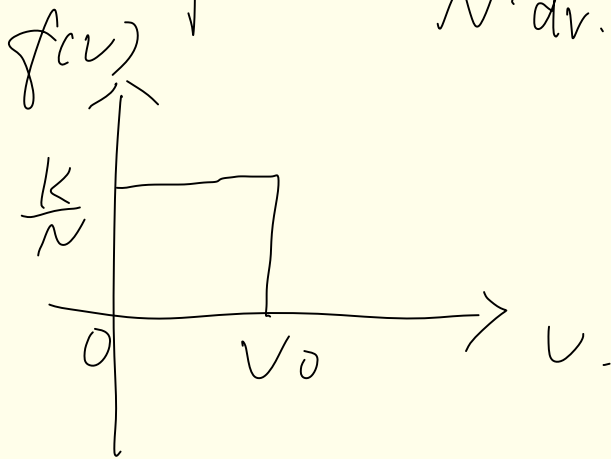
$$\Rightarrow \Delta Q = \frac{3}{4} RT$$

5-14.

解: (1) $dN_v = K dv$ ($V_0 > v > 0, K \neq C$)
 $dN_v = 0$ ($v > V_0$)

$$\therefore \frac{dN_v}{N} = f(v) dv$$

$$\Rightarrow f(v) = \frac{dN_v}{N \cdot dv} = \begin{cases} \frac{K}{N} & (V_0 > v > 0) \\ 0 & (v > V_0) \end{cases}$$



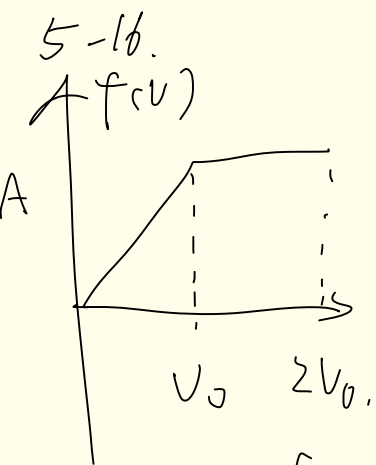
(2),

$$\int_0^{V_0} f(v) dv = 1 \quad (1) \Rightarrow K = \frac{N}{V_0}$$

$$\Rightarrow f(v) = \begin{cases} \frac{1}{V_0} & (0 < v < V_0) \\ 0 & (V_0 < v) \end{cases}$$

$$(3) \bar{v} = \int_0^{\infty} v f(v) dv = \int_0^{V_0} v \cdot \frac{1}{V_0} dv = \frac{V_0}{2}$$

$$\sqrt{\overline{v^2}} = \left(\int_0^{\infty} v^2 f(v) dv \right)^{\frac{1}{2}} = \frac{V_0}{\sqrt{3}}$$



$$\frac{dN}{N} = \begin{cases} \frac{4\pi v^2 A dv}{N} & (0 < v < v_F) \\ 0 & (v_F < v) \end{cases}$$

(1). $f(v) = \frac{4\pi v^2 A}{N}$

$$\int_0^{v_F} \frac{4\pi v^2 A}{N} dv = 1 \Rightarrow A = \frac{3N}{4\pi v_F^3}$$

(2).

$$\begin{aligned} \bar{W} &= \frac{1}{2} m_e \overline{v^2} = \frac{1}{2} m_e \int_0^\infty v^2 f(v) dv \\ &= \frac{1}{2} m_e \int_0^{v_F} \frac{4\pi v^2 A}{N} dv \\ &= \frac{3}{5} \left(\frac{1}{2} m_e v_F^2 \right) = \frac{3}{5} \bar{E}_F \end{aligned}$$